Updated Operations Plan
for
Certificate of Designation
as a Solid Waste Disposal Site
for Tire Derived Fuel (TDF) Use
at the Lyons Cement Plant

Submitted by:

CEMEX, Inc.
Lyons Plant
5134 Ute Highway
P.O. Box 529
Lyons, Colorado 80540

Prepared by:

Banks and Gesso, LLC
720 Kipling Street, Suite 117
Lakewood, Colorado 80215

May 6, 2003

TABLE OF CONTENTS

Page

1.0 Project Introduction and Background 1

1.1 Project Summary 1
1.2 History of Tire Derived Fuel Use at this Site 1
1.3 Request to the State and Boulder County 2

2.0 Facility Information and Contact List 3

3.0 Maps and Site Description 4

3.1 Site Map 4
3.2 Lyons Cement Plant Diagram 4
3.3 Site Description 4

4.0 Description of the Tire Derived Fuel (TDF) Process 5

4.1 Source of Tires 5
4.2 Transportation of Tires 5
4.3 Storage of Tires On-Site 5
4.4 Loading and Burning Process 6
4.5 Cement Kiln Dust (CKD) 7
4.6 Operating Schedule 7
4.7 Inventory Control 7
4.8 Start-Up and Shut-Down Procedures 8
4.9 Personnel and Training 8
4.10 Recordkeeping 8
4.11 Annual Reporting 9

5.0 Fire Protection Plan 10

6.0 Vector Control Plan 11

6.1 Mosquito Control 11

7.0 Closure Plan and Financial Assurance 12

8.0 Other Permits and Approvals 13

8.1 Boulder County Use By Special Review and Site Specific Development Plan 13
8.2 Air Quality Permit 13
8.3 Mining and Reclamation Permit 14

APPENDICES

Appendix A: Tire Trailer Garage Building Description 15

Appendix B: Process Photographs Clinchfield Plant, Georgia 16

Appendix C: Fire Protection Plan 17

Appendix D: Closure Plan 18

1.0 Project Introduction and Background

1.1 Project Summary

The Lyons Cement Plant manufactures portland cement. CEMEX, Inc. proposes to recommence the regular practice of using vehicle tires as a supplemental fuel (Tire Derived Fuel – TDF) in the cement manufacturing process.

Whole used tires will be obtained from Colorado sources, and arrive at the cement plant in tractor trailers. Only clean, dry whole tires will be accepted. Trailers containing tires will be parked in a new garage constructed for this purpose. Tires will remain in the trailers until they are moved to the kiln area and unloaded to a bin and conveying system via hydraulic lift. There will not be any open tire storage or stockpile.

Cemex proposes to substitute approximately 20% tire derived fuel (by BTU value) for fossil fuel (coal and natural gas), resulting in the consumption of up to 4,800 tires per day (~60 tons) or 1,000,000 (one million) tires per year. There will not be any additional noise, dust or odor associated with the use of TDF.

1.2 History of Tire Derived Fuel Use at this Site

The use of tire derived fuel (TDF) at this site was conditionally approved by the Boulder County Board of County Commissioners (BOCC) at a hearing on May 30, 1989, and by Resolution 90-59, signed on February 1, 1990. This Resolution conditionally approved Boulder County Land Use Docket #SU-88-21 ("Lyons Rubber Recycling Facility"), including a Solid Waste Disposal Site

Certificate of Designation (CD) for the purpose of burning used whole and shredded used tires as fuel at the Lyons Cement Plant (formerly known as the Southwestern Portland Cement Company.)

The original Special Use Permit allowed storage of up to 150,000 shredded and 50,000 whole tires in open stockpiles. Tire Derived Fuel (TDF) was allowed as an alternative fuel up to 25% of BTU value.

The Lyons Cement Plant proceeded to burn and store shredded tires as kiln fuel between1989 -1993, at which time the practice was temporarily discontinued for technical and economic reasons.

During its use of TDF in the early 1990's, the Lyons cement plant met all requirements for reporting and stack testing as required by the approved Special Use Permit and Certificate of Designation. Since that time, management has continued to maintain the permits and facilities necessary for TDF use.

1.3 Request to the State and Boulder County

Cemex, Inc. is submitting this Updated Operations Plan to both the Colorado Hazardous Materials and Waste Management Division (the 'Division') and the Board of County Commissioners (BOCC). Cemex requests that the Division conduct a technical review of the Plan and provide a recommendation to the BOCC, and that the BOCC subsequently approve this Updated Operations Plan for Certificate of Designation as a Solid Waste Disposal Facility.

2.0 Facility Information and Contact List

Facility Name: Lyons Cement Plant

Facility Type: Site contains quarries and a portland cement manufacturing facility.

Size/Location: Approximately 792 acres in Sections 21 and 28, Township 3 North, Range 70 West (see 'Site Map')

Site Address: 5134 Ute Highway,
3 miles east of Lyons on Hwy. 66
Boulder County, Colorado

Mailing Address: P.O. Box 529
Lyons, CO 80540-0529

Owner: Cemex, Inc.
1200 Smith Street, Suite 2400
Houston, TX 77002

Site Contacts: Cement Plant Control Room
303-823-2131 phone

Mr. John Lohr, Plant Manager
303-823-2109 ph / 303-823-2199 fax

Mr. Steve Mossberg, Compliance Manager
303-823-2113 ph / 303-823-2199 fax

Mr. Randy Wiley, HazMat Team Leader
303-823-2111 ph / 303-823-2199 fax

3.0 Maps and Site Description

3.1 Site Map

Please see attached map.

3.2 Lyons Cement Plant Diagram

Please see attached map.

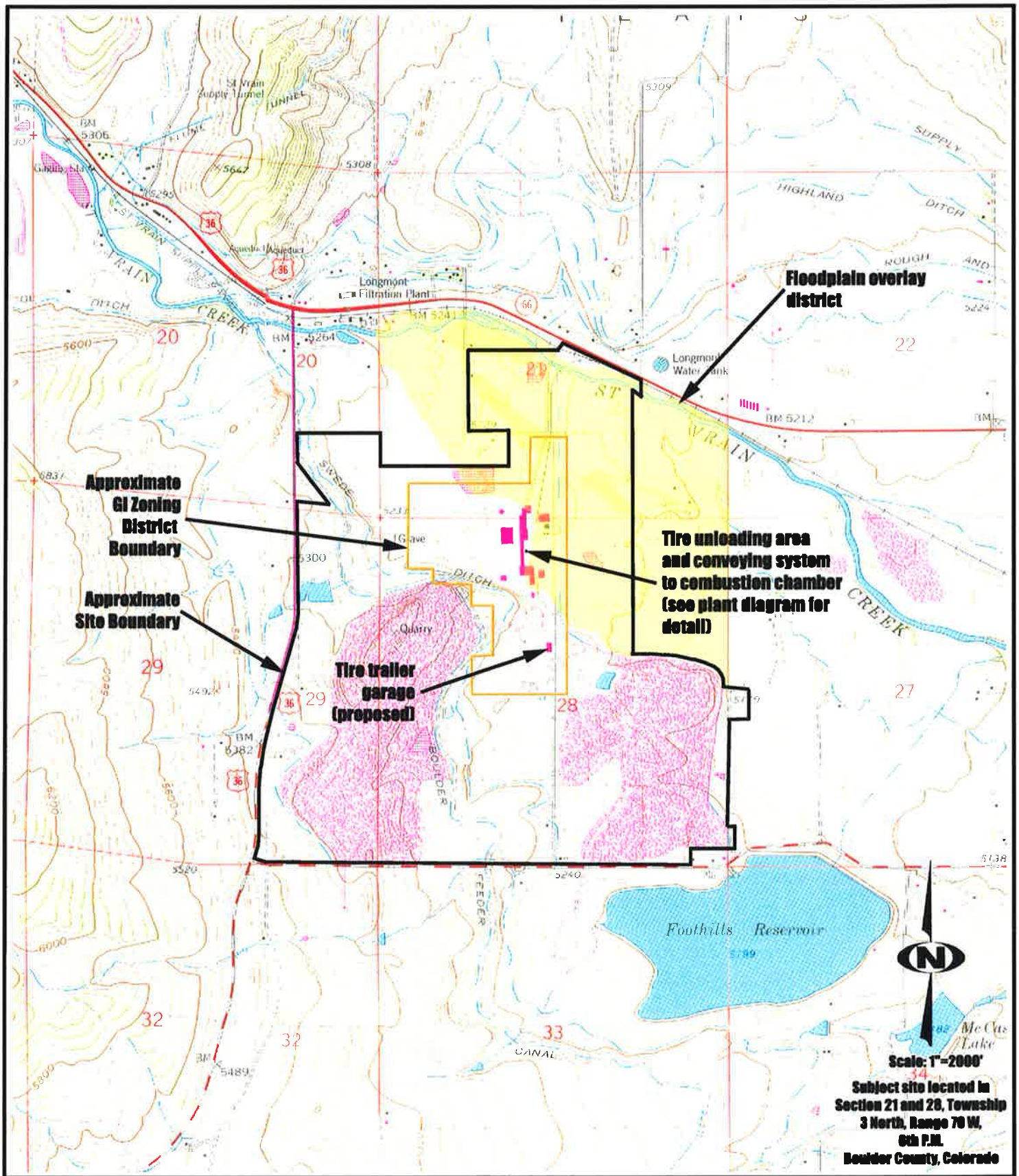
3.3 Site Description

The Lyons cement plant site covers approximately 792 acres south of Highway 66 (see Site Map). Only a small portion of the site (less than 1% of total acreage) will be affected by activities related to tire derived fuel (TDF) use.

TDF use will occur at the cement plant, while trailers containing tires will park in a garage approximately 1000 feet south of the cement plant. Neither of these areas is located within the floodplain district as defined by Boulder County (see Site Map). As an additional precaution, the tire bin will be emptied in the event of a flood threat and rising waters.

All TDF-related activities will occur within the Boulder County General Industrial (GI) zoning district (see Site Map). The GI district is surrounded entirely by Agricultural (A) zoning.

The site has controlled access points, fencing, and 24-hour surveillance to provide security.



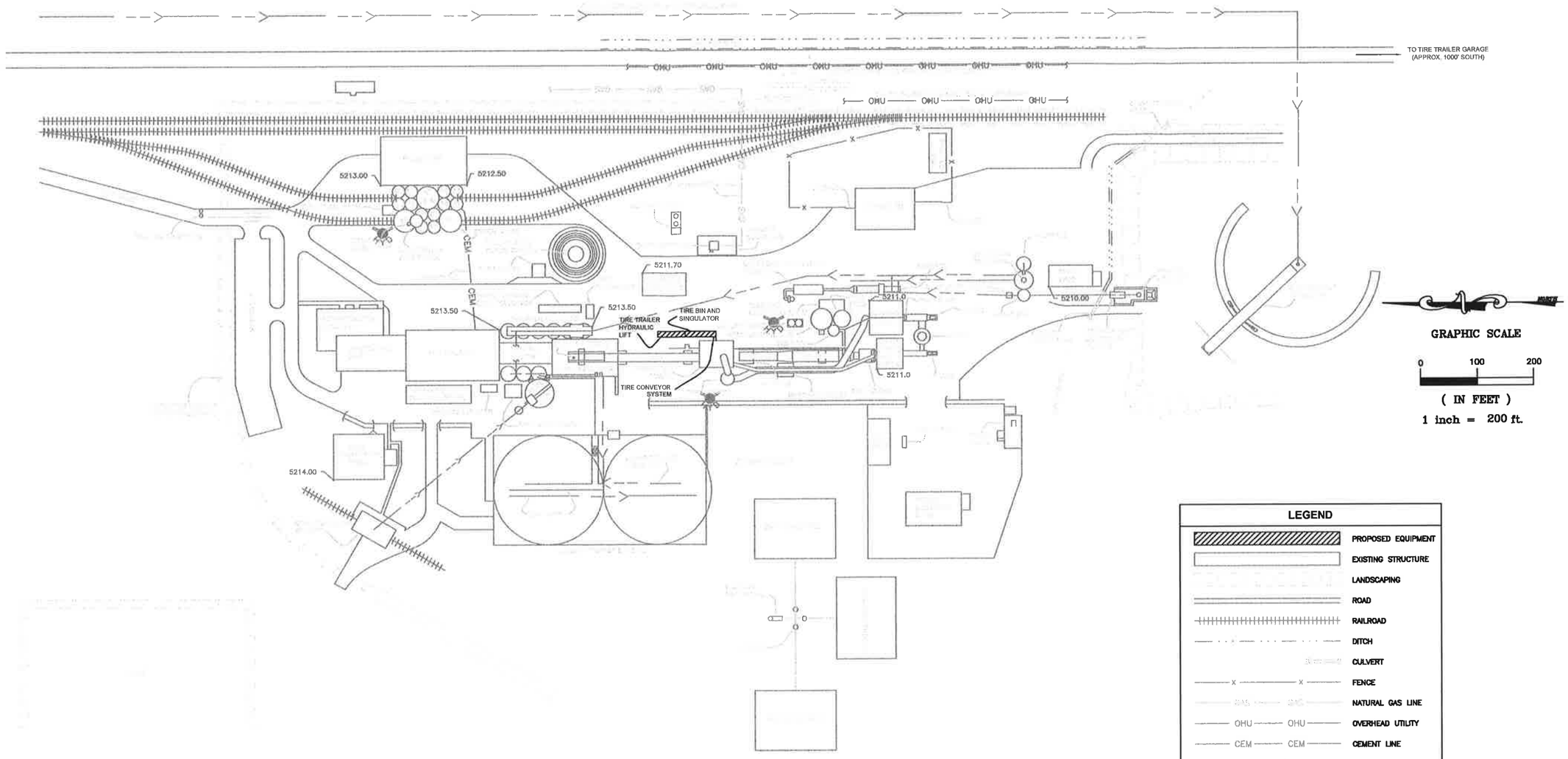
Banks & Gesso, LLC
720 Kipling, Suite 117
Lakewood, CO 80215
banksandgesso.com

Site Map

Cemex, Inc.
Lyons Cement Plant

4/28/03

03021



GRAPHIC SCALE

0 100 200

(IN FEET)

1 inch = 200 ft.

LEGEND	
	PROPOSED EQUIPMENT
	EXISTING STRUCTURE
	LANDSCAPING
	ROAD
	RAILROAD
	DITCH
	CULVERT
x x	FENCE
GAS GAS	NATURAL GAS LINE
OHU OHU	OVERHEAD UTILITY
CEM CEM	CEMENT LINE
	CONVEYOR
	GROUND ELEVATION
	EXISTING FIRE HYDRANT

LYONS CEMENT PLANT DIAGRAM

BANKS AND GESSO, LLC
720 KIPLING ST., SUITE 117
LAKEWOOD, COLORADO 80215
(303) 274-4277

CEMEX, INC.
P.O. BOX 529
5134 UTE HWY.
LYONS, CO 80540

TIRE DERIVED FUEL (TDF)
USE BY SPECIAL REVEIW
BOULDER COUNTY
COLORADO

JOB NO	DATE	SCALE	DRAWN BY	DESIGNED BY	APPROVAL	REV	SHEET
03021	5/1/03	1"=200'	DRF				

4.0 Description of the Tire Derived Fuel (TDF) Process

4.1 Source of Tires

Up to 4,800 tires (or 60 tons, assuming an average weight of 25 pounds per tire) will be delivered and used each day, on average, as a supplemental kiln fuel at the Lyons cement plant. Therefore, up to 1,000,000 whole tires per year will be obtained from local tire dealers or storage sites. All of the tires will be from Colorado sources. It is anticipated that the Lyons Cement Plant will consume approximately 25% of all used tires generated each year in Colorado.

4.2 Transportation of Tires

Approximately 60 tons of clean, dry tires will arrive at the cement plant each day, on average, in 4-6 fully-enclosed single-trailer tractor trailers of various lengths. The tractor trailers will enter the property at the north entrance of the cement plant, and proceed south to the parking garage located approximately 1,000 feet south of the cement plant. The 50 foot wide unpaved haul route is regularly treated with magnesium chloride to control dust.

4.3 Storage of Tires On-Site

A maximum of sixteen 50-foot tire-hauling trailers will park in a new 3-sided, pre-fabricated metal garage located south of the cement plant (see Site Map). The garage will be approximately 60 x 160 feet in dimension, and no more than 25 feet in height. Tires will remain inside the enclosed trailers within the garage until they are moved to the plant area for introduction to the

process. No tires will be stored on the ground outside of trailers. See *Appendix A: Tire Trailer Garage Building Description*.

4.4 Loading and Burning Process

Please see *Appendix B: Process Photographs* for a visual description of the process that is used to move tires from the trailers to the combustion chamber at Cemex's Clinchfield Plant in Georgia. This is a very similar system to the one proposed for the Lyons cement plant.

Whole tires will be moved from the trailer parking garage to the cement plant in the original trailer in which they arrived on site. Trailers will be positioned over an hydraulic lift, which will lift the trailer at an angle such that tires will discharge into the adjacent walking floor bin.

The bin floor conveys the tires to a singulator—a mechanism that removes one tire at a time from the bin at a constant rate. The tires are then conveyed to a chain and hook elevator to take them to the top of the preheater tower. The tires never touch the ground.

At the top of the preheater tower, the tires are conveyed into a double tipping valve (air lock) for controlled insertion into the combustion chamber. Within the combustion chamber, temperatures reach up to 1,800 degrees Fahrenheit, completely incinerating all rubber and steel. Residual ash is incorporated into the cement product and/or captured by the existing dust collection system.

4.5 Cement Kiln Dust (CKD)

Cement Kiln Dust (CKD) is disposed of in an on-site quarry, and this activity is regulated and bonded by the Division of Minerals and Geology (DMG). CDK was sampled and analyzed before and during the November 2002 tire test burn, and test results were submitted to the DMG. No significant chemical difference was found between CKD samples collected before and during the tire test burn.

4.6 Operating Schedule

The cement plant and kiln operate 24 hours a day, 7 days a week. It is expected that tires will be continuously fed to the kiln's combustion chamber as a supplementary fuel source as long as the cement plant is operating. Tire delivery trailers are expected to arrive at the site between 7am and 9pm every day.

4.7 Inventory Control

Tire deliveries will be weighed on certified scales to facilitate all commercial transactions Tire consumption rate will be metered at the cement kiln on a continuous basis. Processed tire weights will be cross-checked with receiving weights to maintain inventory control. Certified static scale weights will be used as the official consumption figure.

4.8 Start-Up and Shut-Down Procedures

The cement kiln will be fitted with tipping valves (air locks) for the input of whole tires. Start-up activities will include the construction of the tire trailer garage, and the commissioning of the tire conveying system. In the event of kiln down-time, tires will remain inside trailers other than those already in the feed bin. It is not expected that tires will remain in the feed bin or on the conveying system for more than 24 hours at a time, except in the case of equipment downtime.

Written notice of anticipated start date will be delivered to the State and Boulder County no earlier than 60 days and no later than 30 days prior to anticipated start-up of tire derived fuel use. Written notice of actual start date will be submitted to the State and Boulder County within 15 days of start-up.

4.9 Personnel and Training

No additional personnel will be added to the Lyons plant payroll as a result of TDF use. There are currently 100 employees on site, not including contract truck drivers. All employees involved with TDF will be given safety, operational and fire prevention training on a regular basis.

4.10 Recordkeeping

The Lyons cement plant maintains daily records of the amount (in tons) and type of each fuel used, including tire derived fuel. Other records to be retained on site associated with TDF use include:

- Daily and annual tire use reports;

- Records showing source of tires;
- Number of incoming tire trailers;
- Downtime and equipment maintenance records;
- Inspection records and follow-up actions;
- Safety and emergency training records;
- Air and cement kiln dust (CKD) monitoring and test data;
- Fire protection plan (see Appendix C);
- Closure plan, including financial assurance (see Appendix D).

These documents will be made available upon request for inspection by Boulder County and the State, and both agencies will be notified whenever new information is added to the operating record.

4.11 Annual Reporting

An annual report will be submitted to the Colorado Department of Public Health and Environment, Hazardous Materials and Waste Management Division by May 1 of every year. This report shall state the amount of tires received and used for fuel on-site for the preceding calendar year.

5.0 Fire Protection Plan

Please see *Appendix C: Fire Protection Plan*.

6.0 Vector Control Plan

6.1 Mosquito Control

The Lyons cement plant will specify that only clean, dry tires be brought on-site in fully enclosed trailers. Tires will be stored inside trailers within a covered and partially enclosed garage. Once tires are discharged from the trailers, they will be consumed for fuel within hours. There will be no open storage of tires, and therefore no opportunity for mosquito larvae accumulations.

7.0 Closure Plan and Financial Assurance

Please see *Appendix D: Closure Plan*.

8.0 Other Permits and Approvals

8.1 Boulder County Use By Special Review and Site Specific Development Plan

Cemex has submitted an application to update Use by Special Review #SU-88-21 ("Lyons Rubber Recycling Facility"), which was conditionally approved on May 30, 1989 by the Boulder Board of County Commissioners (BOCC). The Use by Special Review application includes this Certificate of Designation Updated Operations Plan for the BOCC's approval.

8.2 Air Quality Permit

The Lyons cement plant has continuously maintained air permits allowing the use of tire derived fuels since 1989.

The current Title V Operating Permit (#95OPBO082) was issued by the Colorado Department of Public Health and Environment (CDPHE) Air Pollution Control Division on February 1, 2000, with a revision granted on October 8, 2002. The current permit specifically allows the use of "Whole or Shredded Tires", as well as Natural Gas, Coal and Petroleum Coke, to fuel the rotary kiln (Section 10), at a kiln feed rate of up to 120 tons/hour. Air permit requirements related to the use of tire derived fuel include record keeping of the amount and type of fuel used on a daily basis; meeting emission limitations; and periodic stack testing.

A tire test burn was conducted in November 2002 to measure and compare emissions between use of fossil fuels alone and use of fossil fuels supplemented with tire derived fuel. These reports have previously been submitted to both the State and Boulder County.

8.3 Mining and Reclamation Permit

The Lyons plant site applied for and received Mining and Reclamation Permit #77-208 from the Mined Land Reclamation Board (now called the Colorado Division of Minerals and Geology) in 1977, the same year mining rules and regulations were promulgated by the State. This permit covers mining and processing activities on the subject project.

On-site disposal of Cement Kiln Dust (CKD) is regulated and bonded by the Division of Minerals and Geology (DMG). CDK was sampled and analyzed before and during the November 2002 tire test burn, and test results were submitted to the DMG. No significant chemical difference was found between CKD samples collected before and during the tire test burn.

Appendix A:
Tire Trailer Garage Building Description

Tire Trailer Garage Building Description

The proposed tire trailer garage will be built to provide cover for a maximum of sixteen 50-foot tire-hauling trailers. The garage will be a 3-sided (open to the east), sand or beige colored, pre-fabricated metal building with steel supports. The garage will be approximately 60 x 160 feet in dimension, and no more than 25 feet in height. The proposed garage will be similar in color and building materials to the Dowe Flats crusher building, shown in the photo below.

Photograph of the Dowe Flats crusher building, located north of Highway 66. The proposed garage will look architecturally similar to this building (similar color and building materials), but will differ in height and size.



Appendix B:
Process Photographs
Clinchfield Plant, Georgia


Trailer adjacent to hydraulic lift.
1


Looking into bin from lift.
2


Unloading tires from trailer to walking floor bin
3


Walking floor bin with tire singulator at far end.
4


Tires coming out of singulator and being conveyed to preheater tower
5


Tires entering preheater structure and being moved to top of structure via chain and hook elevator.
6


CEMEX
Chain and hook elevator taking tires to combustion chamber at top of preheater tower.
7


CEMEX
Tires being discharge from elevator onto conveyor and into double tipping valve for insertion into combustion chamber.
8


CEMEX
Combustion chamber where tires are burned as a supplemental fuel to heat the cement kiln.
9

Appendix C:
Fire Protection Plan

Fire Protection Plan
for
Tire Derived Fuel (TDF) Facilities
Lyons Cement Plant, CO

Submitted by:

CEMEX, Inc.
Lyons Plant
5134 Ute Highway
P.O. Box 529
Lyons, Colorado 80540

Prepared by:

Banks and Gesso, LLC
720 Kipling Street, Suite 117
Lakewood, Colorado 80215

May 6, 2003

TABLE OF CONTENTS

Page

1.0 Facility Information and Contact List 1

2.0 Local Fire Department Contacts 2

3.0 Maps 3

3.1 Site Map 3
3.2 Cement Plant Diagram 3

4.0 Fire Protection Plan 4

4.1 Cement Plant Fire Plan 4
4.2 Tire Trailer Garage Fire Plan 4
4.3 Fire Notification 5

APPENDICES

Appendix A: Hygiene Fire Department Approval Letter 6

Appendix B: Hot Work Permit 8

1.0 Facility Information and Contact List

Facility Name:	Lyons Cement Plant
Facility Type:	Portland cement manufacturing facility burning used tires as fuel in the cement kiln.
Site Address:	5134 Ute Highway, 3 miles east of Lyons on Hwy. 66 Boulder County, Colorado
Mailing Address:	P.O. Box 529 Lyons, CO 80540-0529
Owner:	Cemex, Inc. 1200 Smith Street, Suite 2400 Houston, TX 77002
Site Contacts:	Cement Plant Control Room 303-823-2131 phone
	Mr. John Lohr, Plant Manager 303-823-2109 ph / 303-823-2199 fax
	Mr. Steve Mossberg, Compliance Manager 303-823-2113 ph / 303-823-2199 fax
	Mr. Randy Wiley, HazMat Team Leader 303-823-2111 ph / 303-823-2199 fax

2.0 Local Fire Department Contacts

First Response:

Hygiene Volunteer Fire Department
303-776-2950
P.O. Box 83
Hygiene, CO 80533

Chief: John (J.J.) Hoffman
303-775-1461 cell

Back-Up:

Lyons Fire Protection District
303-823-6611
251 Broadway Avenue
Lyons, CO 80540

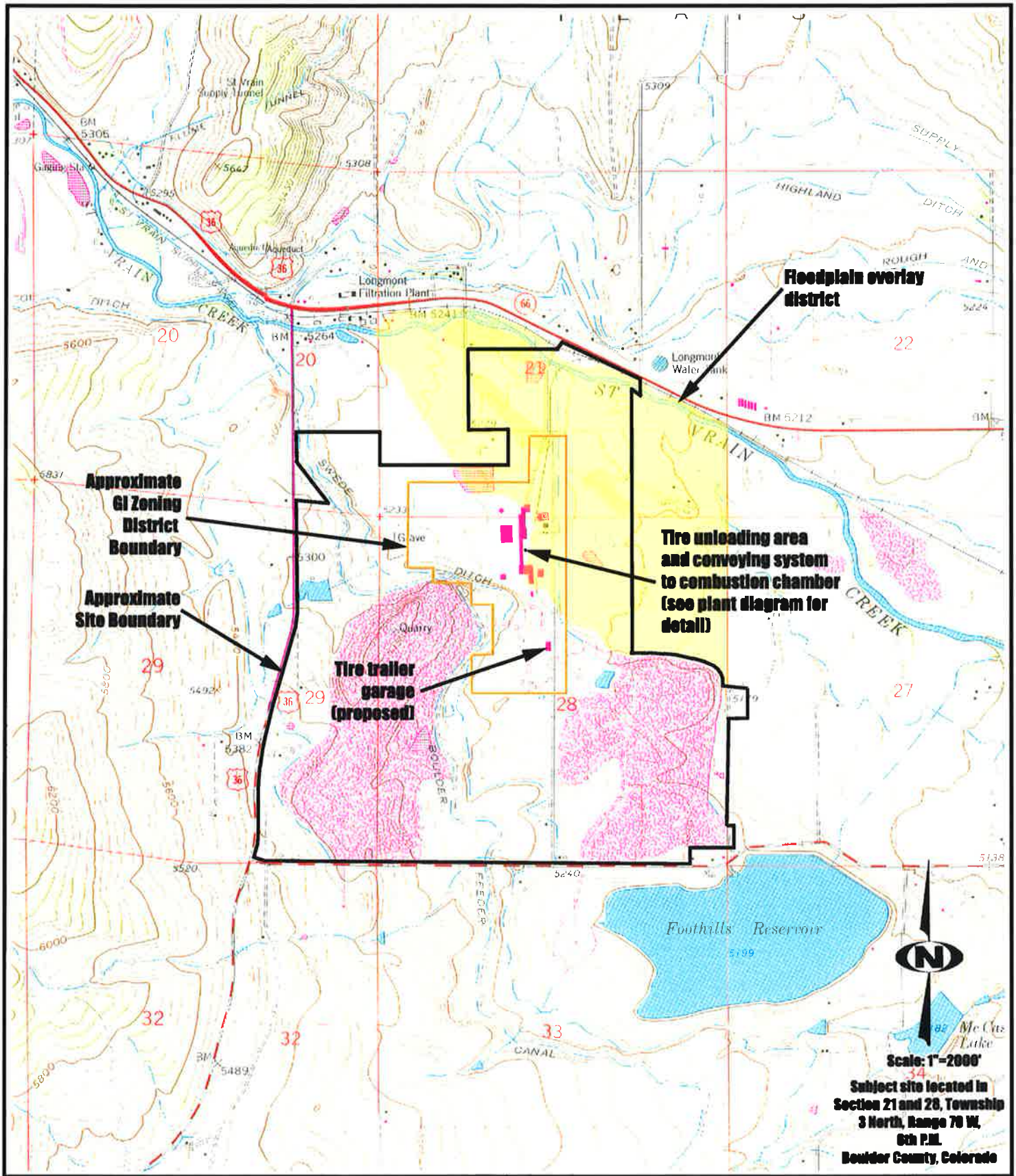
3.0 Maps

3.1 Site Map

The attached site map shows the location of the tire trailer storage garage.

3.2 Cement Plant Diagram

The attached diagram shows the location of the proposed tire bin relative to the existing cement plant, and the approximate location of the existing fire hydrants (3).



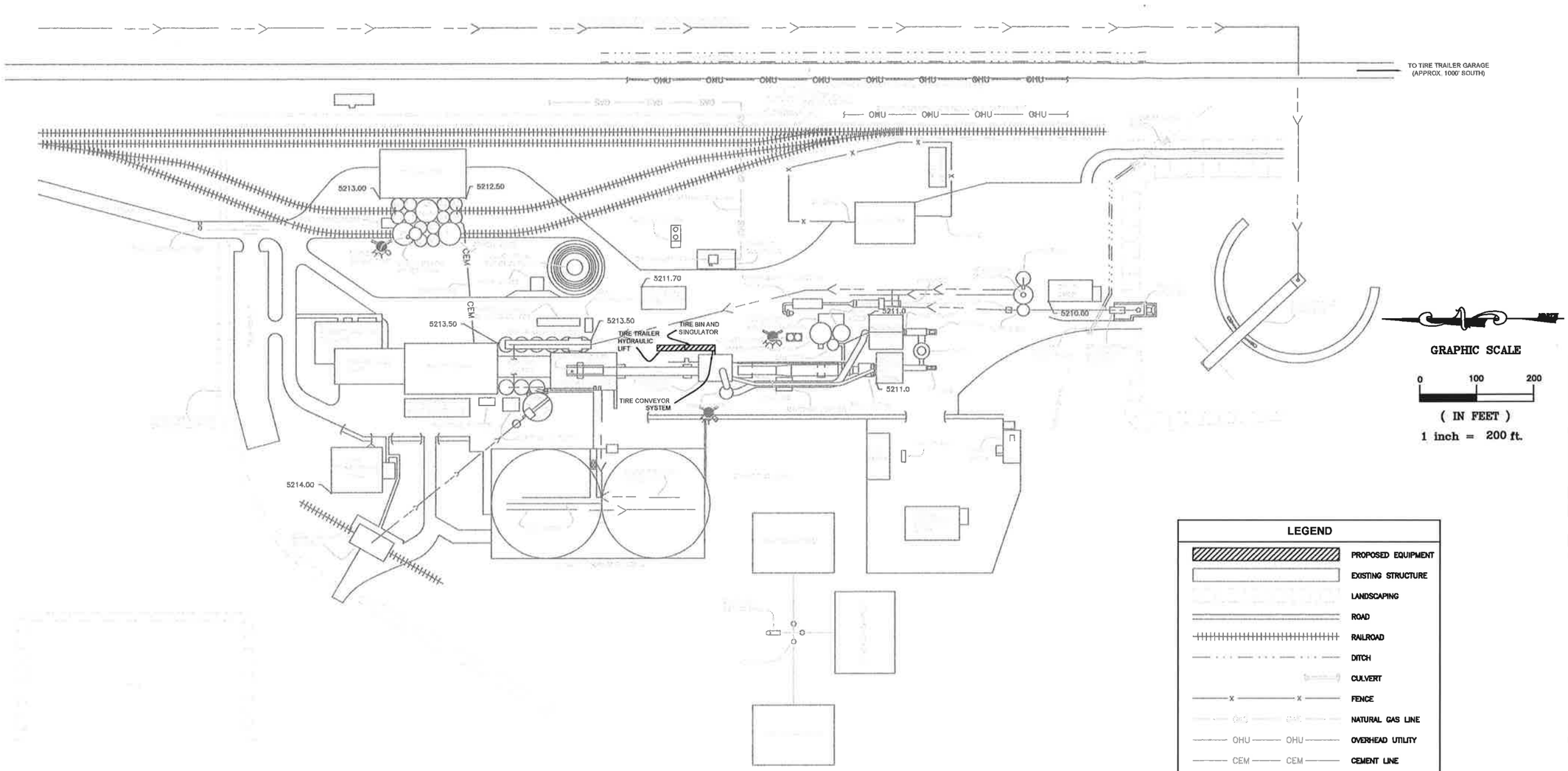
Banks & Gesso, LLC
720 Kipling, Suite 117
Lakewood, CO 80215
banksandgesso.com

Site Map

Cemex, Inc.
Lyons Cement Plant

4/28/03

03021



GRAPHIC SCALE

0 100 200

(IN FEET)

1 inch = 200 ft.

LEGEND

- PROPOSED EQUIPMENT
- EXISTING STRUCTURE
- LANDSCAPING
- ROAD
- RAILROAD
- DITCH
- CULVERT
- FENCE
- NATURAL GAS LINE
- OVERHEAD UTILITY
- CEMENT LINE
- CONVEYOR
- GROUND ELEVATION
- EXISTING FIRE HYDRANT

LYONS CEMENT PLANT DIAGRAM

BANKS AND GESSO, LLC
720 KIPLING ST., SUITE 117
LAKEWOOD, COLORADO 80215
(303) 274-4277

CEMEX, INC.
P.O. BOX 529
5134 UTE HWY.
LYONS, CO 80540

TIRE DERIVED FUEL (TDF)
USE BY SPECIAL REVEIW
BOULDER COUNTY
COLORADO

JOB NO	DATE	SCALE	DRAWN BY	DESIGNED BY	APPROVAL	REV	SHEET
03021	5/1/03	1"=200'	DRF				

4.0 Fire Protection Plan

4.1 Cement Plant Fire Plan

Dry chemical extinguishers will be placed at the tire bin, conveyor, and elevator areas. A hydrant is located approximately 50 feet west of the preheater tower. See 'Cement Plant Diagram' for proposed location of trailer lift and tire bin, and locations of three existing hydrants.

4.2 Tire Trailer Garage Fire Plan

Up to 16 trailers will park in the tire trailer garage, holding a maximum total of approximately 300 tons, or 24,000 tires (assuming 80 tires per ton). The tires will be stored in enclosed trailers which will minimize any exposure or risk of combustion.

A standard fire hydrant will be installed within 50 yards of the garage[1]. This hydrant will be hooked to the plant process water line. Dry chemical extinguishers will also be placed at the tire trailer garage.

A secondary water source (lake) is located approximately 1,800 feet from the proposed location of the garage (see 'Site Map').

Individual trailers will normally not be on the premises for more than one week at a time. Exceptions to this would be in the case of TDF equipment breakdown or kiln downtime.

Any welding or hot work on TDF storage or handling equipment will require a "Hot Work" permit to identify potential risks and remediation (see Appendix B).

[1] Per Hygiene Fire Department recommendation (see Appendix A).

Hygiene Fire District representatives will be oriented to the new facility prior to commissioning.

4.3 Fire Notification

In the event of a fire, Lyons cement plant personnel will immediately call 911, notify the Hygiene Fire Department, and act to contain the fire. The local health department and the Colorado Department of Public Health and Environment (CDPHE), Hazardous Materials and Waste Management Division will also be notified. Within two weeks of this notification, a report shall be submitted to Boulder County and CDPHE.

This report shall describe the origins of the fire, the actions that have been taken, current and/or planned actions, expected results from these actions, and an approximate date of resolution of the problems generated by the fire.

Appendix A:
Hygiene Fire Department Approval Letter

Hygiene Fire Department
P.O. Box 83
Hygiene, CO 80533

April 25, 2003

Cemex
Attn: John Lohr
Plant Manager

Dear John,

After our meeting on April 24, 2003 I review your proposal for a Fire Protection Plan for the Tire Derived Fuel Facilities for the Lyons Cement Plant, and found that the proposed plan is a good plan. The tire storage area is located in a very good location because there are no other structures in a close proximity in the event of a fire and there is also a large amount of water supply locations very close with easy access. As per our conversation we would like to see a hydrant installed off the main line pump system in the intersection just North of the storage area approx 50 yards away.

I find no problems with the proposed plan and feel that it is a very reasonable emergency response plan for this facility.

Please feel free to call me with any questions regarding this situation.

Sincerely,

John Hoffman (J.J.)
Chief Hygiene Fire Department

Appendix B: Hot Work Permit

PRE-TASK CHECKLIST (Hot Work Permit)

Work Order # _____	**Date** _____

EXPLAIN TYPE OF WORK

Combustion Hazards	Spark Producing Hazards	Other Hazard
Welding _____	Chipping _____	Electrical _____
Burning _____	Grinding _____	Vehicle Entry _____
Open Plant _____	Drilling _____	Other _____

Check type of hazard

Toxic	Spilled	Tightening joints	Radioactive equipment
Corrosive	Hot	Under pressure	Welding under pressure
Flammable	Chemical cleaning	Vessel entry	Welding on vehicle
		Unguarded equipment	Unplugged lines/equipment
		Other	

Check special procedures required

PRIOR TO BEGINNING WORK: (check off as completed)

___ Lockout – disable equipment / isolate energy sources. (list below)

___ Insure that all necessary PPE is worn (list below).

____ Inspect machine and work area for hazards (list below).

Personal safety preparation

- Protect against _____
- Air mask
- Dust respirator
- Coveralls
- Safety shower
- Face shield
- Chemical / splash goggles
- Radio
- Evacuation instructions
- Two escape routes
- Life line harness / life line
- Hand protection
- Foot protection
- Nomex suits
- Added assistance
- Other _____

Equipment safety preparations

ISOLATE EQUIPMENT	ELECTRICAL SAFETY
Stop transfers	Lock circuits
Lock-out and tag-out	Explosion proof equipment
Disconnect and blank	Non sparking tools
Post work signs	Spark containers
Set up barricade/caution tape	Welding arc spark protector
Block roadway	Grounding
Other _____	Other _____
CLEAN EQUIPMENT	**FIRE SAFETY**
Fill and drain	Bleed steam hose
Remove deposits	Run water hose
Water wash	Fire watch
Inspect for leaks	Keep area wet
Visually inspect	Cover sewer openings
Ventilate	Fire extinguishers
Other _____	Fire blanket

ENVIRONMENTAL: All fluids and waste products must be disposed of properly.

HOUSEKEEPING: (check off as completed)

___ Clean up work place after completion.

___ Clean and return tools to storage.

___ Dispose of worn parts, dirt and debris.

___ Properly store unused parts and supplies.

Signature(s) _____ _____ _____

_____ _____ _____

Appendix D:
Closure Plan

Closure Plan
for
Tire Derived Fuel (TDF) Facilities
Lyons Cement Plant, CO

Submitted by:

CEMEX, Inc.
Lyons Plant
5134 Ute Highway
P.O. Box 529
Lyons, Colorado 80540

Prepared by:

Banks and Gesso, LLC
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May 6, 2003

TABLE OF CONTENTS

Page

1.0 Introduction and History 1

2.0 Facility Description 2

3.0 Waste Description and Amount 2

4.0 Equipment Closure 3

5.0 Tire Trailer Garage Closure 3

6.0 Safety 3

7.0 Post-Closure Care 4

8.0 Closure Notification 4

9.0 Closure Certification 4

10.0 Closure Schedule 5

11.0 Closure Cost 5

12.0 Financial Assurance 5

1.0 Introduction and History

Cemex, Inc. operates quarries and a portland cement plant in Lyons, Colorado. The Lyons cement plant includes a rotary cement kiln which can be fueled, in part, by the addition of whole tires into the combustion chamber in the preheater tower.

The use of tire derived fuel (TDF) at this site was conditionally approved by the Boulder County Board of County Commissioners (BOCC) at a hearing on May 30, 1989, and in Resolution 90-59, signed on February 1, 1990. This Resolution conditionally approved Boulder County Land Use Docket #SU-88-21 ("Lyons Rubber Recycling Facility"), including a Solid Waste Disposal Site Certificate of Designation (CD) for the purpose of burning used whole and shredded used tires as fuel at the Lyons Cement Plant.

The Lyons plant proceeded to burn and store whole and shredded tires on site between 1989 and 1993. A Closure Plan was prepared in March 1995 in compliance with solid waste regulations. The 1995 Plan addressed the potential closure of shredded and whole tire storage piles.

Since the original Closure Plan was prepared, changes have been made to the way tires are expected to be stored and delivered to the kiln.

This new Closure Plan is part of an Updated Operations Plan that Cemex, Inc. is submitting to the Colorado Department of Public Health and Environment (CDPHE), Hazardous Materials and Waste Management Division as required by solid waste regulations.

2.0 Facility Description

Whole, clean, dry tires will be obtained from Colorado sources and brought on site by standard, enclosed tractor trailers of various sizes contracted for this purpose. The facility will not be open to the general public for disposal of used tires, except for a special community clean-up day once each year.

The use of tire derived fuel (TDF) and related activities involves only a small portion of the site, and its use is incidental to the facility's primary purpose of manufacturing cement. Closure of TDF-related activities might occur prior to or concurrently with the closure of the cement plant itself.

TDF equipment and facilities subject to this closure plan include:

- Tire trailer garage;
- Trailer hydraulic lift;
- Tire bin with singulator;
- Tire conveyor system; and
- Combustion chamber input valves.

The tire trailer garage is located approximately 1,000 feet south of the cement plant. All other TDF equipment is located at the cement plant, primarily adjacent to and east of the preheater tower.

3.0 Waste Description and Amount

Only whole clean, dry used tires are brought on site for use as tire derived fuel (TDF). A maximum of approximately 300 tons of tires (24,000 tires, assuming 80 tires per ton) will be on-site at any given time, including tires located in trailers, and those in the bin and on the conveying system.

4.0 Equipment Closure

Tire derived fuel (TDF) equipment includes an hydraulic lift, walking floor bin with singulator, conveyors and elevator, and valves for insertion of tires into the combustion chamber in the preheater tower. Closure of the TDF facilities would include the removal of all tires located on this equipment, and the dismantling of the equipment itself. All of the equipment associated with TDF use is mobile and easily dismantled. The plant area will be visually inspected upon closure for any remaining tires or residual debris.

5.0 Tire Trailer Garage Closure

The tire trailer garage will be able to store a maximum of sixteen 50-foot long trailers. Tires will remain inside the trailers until they are moved to the cement plant area and unloaded into the bin. Closure of the tire trailer garage would require only the removal of all loaded trailers, and visual inspection of the garage and surrounding area for any remaining tires and/or residual debris.

6.0 Safety

All personnel performing activities associated with the closure of the garage and tire derived fuel related equipment will be provided with the appropriate safety training and equipment. All closure activities will be performed in accordance with health and safety standards specified in the Mine Safety and Health Administration (MSHA) regulations (30 CFR Parts 1 through 199).

7.0 Post-Closure Care

No post-closure activities will be required, since all remaining tires and associated equipment will be removed from the site during closure.

8.0 Closure Notification

At least one hundred twenty (120) days prior to the permanent closure of tire derived fuel (TDF) activities, Cemex will notify, in writing, Boulder County, and the Colorado Hazardous Materials and Solid Waste Management Division, and will notify the general public by placing signs of suitable size at the facility entrances.

9.0 Closure Certification

Within sixty (60) days of the completion of final closure of TDF activities, Cemex, Inc. will submit to the State and Boulder County, a certification and report that the TDF equipment and facilities have been closed in accordance with the specifications in the approved Closure Plan. The certification and report will be signed by a Colorado registered professional engineer. The report will include a summary of the closure activities performed, including the following:

- Copies of any pictures taken during closure;
- Copies of daily logs; and
- Results of visual inspections.

10.0 Closure Schedule

The removal of any remaining tires and loaded tire trailers, and the dismantling and removal of tire derived fuel (TDF) related equipment, will be completed within sixty (60) days of commencing closure activities.

11.0 Closure Cost

Labor to remove remaining tires from bin and conveying system:	
12 hours at $15 per hour =	$180
Labor to dismantle TDF equipment:	
200 hours at $20 per hour =	$4,000
Supervisory cost:	
40 hours at $35 per hour =	$1,400
Transportation cost to remove up to 20 trailers:	$4,000
Disposal fee for 24,000 tires:	$7,500
Removal cost for dismantled equipment:	$10,000
Professional Engineer Certification/Report:	$5,000
TOTAL CLOSURE ESTIMATE:	**$32,080**

12.0 Financial Assurance

Cemex, Inc. will maintain financial assurance, as required by the State, to cover the removal of any stored tires in the event that TDF activities are permanently discontinued.